Homework 3 Solutions: Reading Between the Lines: Predicting Supreme Court Votes Based on Oral Argument Emotions

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Acknowledgments

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1 Preliminaries

Q1: Write a concise, 3-sentence description of the dataset as if you were explaining it to a colleague. What kind of information does each row represent?
A: This is a case-level dataset about judicial cases from the American Supreme Court. It provides rich information about the justices' emotional responses during oral argument. In particular, the data captures their voice pitch, enabling analysis of their emotional state toward the petitioners and respondents.
Q2: What is the unit of observation in this dataset? In other words, what does each observation (or row) correspond to?
A: Each observation (row) is a case i judged by a justice y in the Supreme Court in a particulat year z.
Q3: What does it mean when a Supreme Court justice votes in favor of the petitioner? Why is accurately predicting this decision important for policy and legal outcomes? (Hint: Refer to the authors' article for insights.)
A: It means the Supreme Court believes that the petition was ruled incorrectly in lower courts. The reasons may vary. Accurately predicting the decisions is important mainly because it reduces uncertainty. Following the author's articles, some decisions might take months to be made or released. A model prediction can help stakeholders anticipate the likely outcome and prepare the next steps accordingly.

Q4: How many missing values are there in the dataset? Identify which variables have missing data and explain why this might matter for your analysis.
A: There are 232 missing values (5,568 obs., ~4.5% of the observations) in 24 of the 35 variables.

2 Getting to Know the Data and Descriptive Statistics

Q5: Describe the general distribution of the variables petitioner vote, pitch diff, and petitioner harvard pos. Are there any patterns in how these values are spread out? Do you notice any missing data or extreme values that might be important for analysis?
A: *pitch_diff* varies from -20 to nearly 4. Its average is ~ -0.074 ($sd = 0.889$), and its IQR is ~ 0.873, which shows that most of the observations don't vary too much. This also indicates that the extreme values, such as -20 and 4, are likely outliers. There are no missing values, which is good, as this is our variable of interest.

petitioner_harvard_pos is a count variable with an average of ~ 9 ($sd = 8.778$) and an IQR $= 10$. Hence, most observations are concentrated within that range. The maximum value is 90, indicating a likely right-skewed long tail and an outlier.

petitioner_vote has a mean of 0.545 ($sd = 0.498$), indicating that petitioner-favoring votes are roughly evenly split in the sample.

Q6: The dataset includes variables based on justices' speech patterns and word choice during oral arguments. How was pitch diff measured, and why was it standardized? How was petitioner harvard pos created, and what are some potential challenges of using a predefined list of positive words? Could this approach misinterpret tone or context, such as when a justice uses sarcasm?
A: pitch_diff was measured by subtracting the vocal pitch in questions toward the petitioners from the vocal pitch in questions directed toward the respondent. It was standardized to account for systematic differences between Justices (gender, age) and to any measurement error associated with the speech synthesis program, Praat.

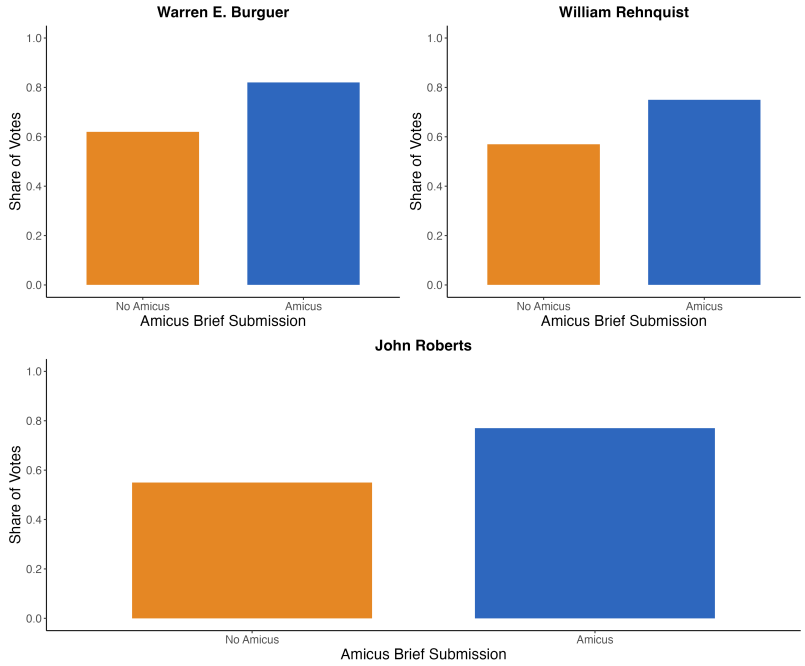
petitioner_harvard_pos was created based on a predefined list of positive words. The sole number of posititve words might be very misleading, because i) the Justice could use a negation before the word, i.e. 'that's not compassionate from your client'. Compassionate will be counted as a positive word toward the petitioner, but the context is undoubtedly negative; ii) as the prompt says, it does not account for sarcasm, analogies, or even comparisons from other cases. The Justice can relate the current case to another case in which the petitioner was right, thereby saying something good about a past petitioner. Again, one more positive word toward the petitioner, but it's not for them. Overall, the approach has a limitation: it struggles to account for the context of word choice.

Q7: The petitioner harvard pos variable is based on a predefined dictionary of positive words. What are some ways this method might introduce errors? How could these errors impact the interpretation of the results?
A: Considering the impact of errors, we might find a very strong and positive relationship between voting in favor of the petitioner and the number of positive words directed toward them. For reasons described above, if we account for the context of the choice of words, that relationship might be less strong or even nonexistent. The prediction might be flawed.

Q8: The bar chart compares the proportion of votes in favor of the petitioner across three chief justices, depending on whether the Solicitor General submitted an amicus brief. What trends do you observe in how each justice responded to amicus support? What factors might explain differences in their voting patterns?

A: Across all chief justices, the share of votes in favor of the petitioner is higher when there is an amicus brief submission. The difference is around 33%, on average. The Solicitor General represents the interests of the U.S. government in a particular case. They are allowed to give their opinion on the matter, and this might influence the justice's decision. We may assume that in our data they were usually on the petitioner's side, but that is not always the case.

Figure 1: Share of votes in favor of the petitioner, by amicus brief submission

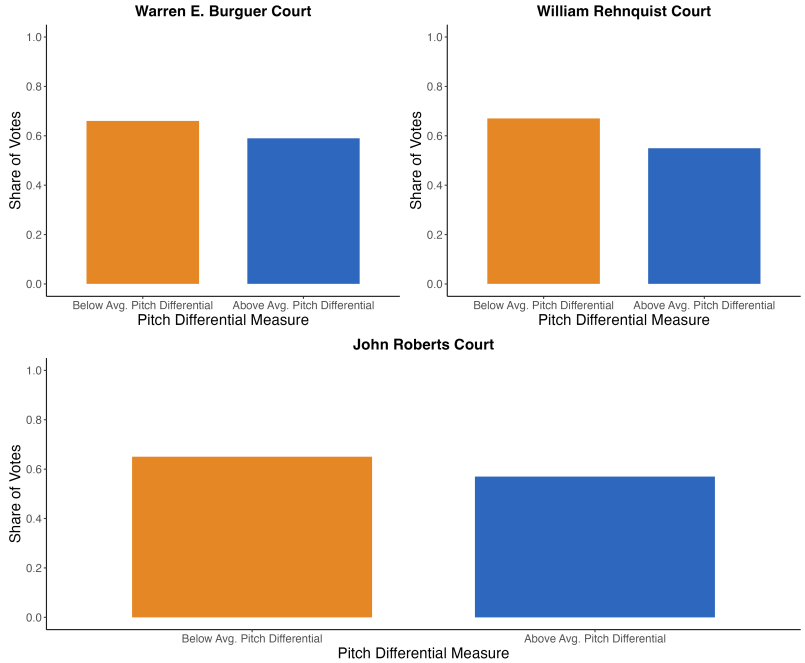


Notes: This figure shows the share of votes in favor of the petitioner across three chief justices, depending on whether the Solicitor General submitted an amicus brief (blue bar; orange bar if otherwise). There are 70 cases where there is no amicus brief information. N = 618.

Q9: The second bar chart examines how pitch differences in justices' speech relate to their votes across different court periods. What does the chart suggest about the relationship between pitch and voting behavior during the Burger, Rehnquist, and Roberts courts? Do you see any shifts in this pattern over time?

A: Across all court periods, we see the same pattern: a negative relationship between pitch differences and the justices' voting behavior. That is, whenever the pitch difference is higher than the sample average, it is less likely that the justice voted in favor of the petitioner. In other words, when the justices change their voice pitch more than usual, indicating more emotional traits, it's more likely they will vote against the petitioner.

Figure 2: Share of votes in favor of the petitioner, by court period and pitch differences



Notes: This figure shows the share of votes in favor of the petitioner across three court periods, depending on whether the pitch difference was above its average sample value (blue bar; orange bar if otherwise). There are 70 cases where there is no amicus brief information. N = 618.

3 Regression Analyses

Q10: What do you learn from the first regression (predicting petitioner vote using pitch diff and pr petitioner pos)? How do these predictors relate to justices' voting behavior?

A: I learn that there's a negative association between voting in favor of the petitioner and the justice's pitch difference. In particular, a one-standard-deviation increase in the pitch difference is associated with an average decrease of 5.7 percentage points in the predicted probability of voting in favor of the petitioner, all else constant. The effect is significant at 1%.

We find no statistically significant effect of the difference in the proportion of positive words used when addressing the petitioner versus the respondent. This indicates that the fact that the Justice might have used more positive words towards one than the other does not influence the likelihood of them voting in favor of the petitioner.

Q11: Why is it important to control for justice-specific effects when running the regression? How do the results change when these controls are added?

A: Justice fixed effects are important because each Justice may systematically differ in how they engage with petitioners and respondents, including in their vocal pitch patterns. By including Justice fixed effects, the model accounts for idiosyncratic differences in judicial style, temperament, and overall propensity to vote in favor of the petitioner.

For some Justices, the probability of voting in favor of the petitioner decreases more than in the baseline model. For example, it decreases, on average, by an additional 11.2 percentage points if the case was led by Justice JP Stevens (significant at the 1% level), all else held constant. The pitch

Table 1: Regression Results (Voting in Favor of the Petitioner)

	Dependent variable:		
	Favor Petitioner Vote		
	(1)	(2)	(3)
Pitch Difference	-0.057***	-0.058***	-0.058***
	(0.008)	(0.008)	(0.008)
Positive Words Difference	0.041	0.057	0.058
	(0.160)	(0.160)	(0.160)
Justice: AScalia		-0.059**	-0.059**
		(0.027)	(0.027)
Justice: BRWhite		-0.155	-0.154
		(0.166)	(0.166)
Justice: CThomas		-0.608	-0.608
		(0.494)	(0.494)
Justice: DHSouter		-0.039	-0.039
		(0.032)	(0.032)
Justice: EKagan		-0.029	-0.029
		(0.051)	(0.052)
Justice: HABlackmun		-0.492**	-0.492**
		(0.249)	(0.249)
Justice: JGRoberts		0.006	0.006
		(0.035)	(0.036)
Justice: JPStevens		-0.112***	-0.113***
		(0.032)	(0.032)
Justice: LFPowell		-0.337	-0.354
		(0.248)	(0.251)
Justice: RBGinsburg		-0.077***	-0.077***
		(0.028)	(0.028)
Justice: SAAlito		-0.029	-0.029
		(0.040)	(0.040)
Justice: SDOConnor		0.048	0.046
		(0.035)	(0.036)
Justice: SGBreyer		-0.075***	-0.075***
		(0.029)	(0.029)
Justice: SSotomayor		-0.098**	-0.098**
		(0.042)	(0.043)
Justice: TMarshall		-0.175	-0.201
		(0.144)	(0.156)
Justice: WEBurguer		-0.015	-0.049
		(0.080)	(0.111)
Justice: WHRehnquist		0.019	0.018
		(0.037)	(0.038)
Court Period: Rehnquist			-0.034
			(0.078)
Court Period: Roberts			-0.034
			(0.079)
Constant	0.544***	0.589***	0.623***
	(0.007)	(0.021)	(0.081)
Observations	4,977	4,977	4,977
R^2	0.010	0.020	0.020
Adjusted R^2	0.010	0.016	0.016
Residual Std. Error	0.495 (df = 4974)	0.494 (df = 4957)	0.494 (df = 4955)
F Statistic	26.305*** (df = 2; 4974)	5.258*** (df = 19; 4957)	4.764*** (df = 21; 4955)

Note: This table shows the results from the multivariate regression estimation. The dependent variable is an indicator of whether the Supreme Court Justice voting in favor of the petitioner (1, 0 otherwise). The independent variable of interest is the Justice oral pitch difference, that is, the difference between the pitch in questions directed toward petitioners and the pitch in questions directed toward respondents. The vocal pitch was converted to standard deviations above and below the Justices' average vocal pitch. Justice-specific and court period effects included. *p<0.1; **p<0.05; ***p<0.01.

difference estimate, however, changes only slightly from -.057 to -.058 (still statistically significant at 1% level).

Q12: How does adding court term indicators change the model's explanatory power? What does the adjusted R2 tell us?

A: Relative to the baseline model, adding court term indicators changes the model's explanatory power from 0.01 to 0.02. R^2 remains very low, indicating that only 2% of the variation in the outcome is explained by the independent variables' variation. The adjusted R^2 falls to 0.016 (hence, only 1.6% of the variation in the outcome is explained by the explanatory variables' variation), suggesting that the R^2 overestimated the model's explanatory power after adding court term indicators.

Q13: In the interaction model for pitch diff and court period, what patterns do you observe? Does the relationship between vocal pitch and voting behavior change across different Supreme Court periods?

A: The figure supports our finding that pitch difference negatively predicts voting behavior, but the court period indicator does not appear to meaningfully differentiate that relationship. The overlapping CIs suggest that the interaction terms for court period are likely not statistically significant. However, when we look at the regression results, we do find that relative to the Burger Court baseline, a one standard deviation increase in pitch difference is associated with an average decrease in the predicted probability of voting for the petitioner of 7.1 percentage points during the Rehnquist period and 5.5 percentage points during the Roberts period, all else held constant. These interaction effects are statistically significant, though at more modest levels than the main effect (5% and 10%, respectively).

This loss in statistical significance was expected, following Figure 3. It's important to note, however, that if we disregard the area where pitch_difference has extreme values (pitch_diff < -1), Renhnquist and Roberts court periods' CIs overlap less with Burguer's CI. This could help explain the regression results in Table 2: significant but not *too* significant.

Q14: Compare the six progressively built regression models. What do you learn from the addition of each new predictor or interaction term? How does controlling for these factors refine our understanding of justices' voting behavior?

A: Three main findings emerge from Table 2. First, the coefficient on pitch difference remains stable across models (around -0.055 to -0.057 percentage points), which suggests the estimate is robust to the inclusion of additional controls. The exception is Model 5, where the introduction of interaction terms changes the interpretation of the baseline coefficient.

Second, Model 5 reveals that the negative relationship between pitch difference and voting behavior is more pronounced in later court periods. Relative to the Burger Court baseline, a one standard deviation increase in pitch difference is associated with an average decrease in the predicted probability of voting for the petitioner of 7.1 percentage points during the Rehnquist period and 5.5 percentage points during the Roberts period, all else held constant. These interaction effects are statistically significant, though at more modest levels than the main effect (5% and 10%, respectively).

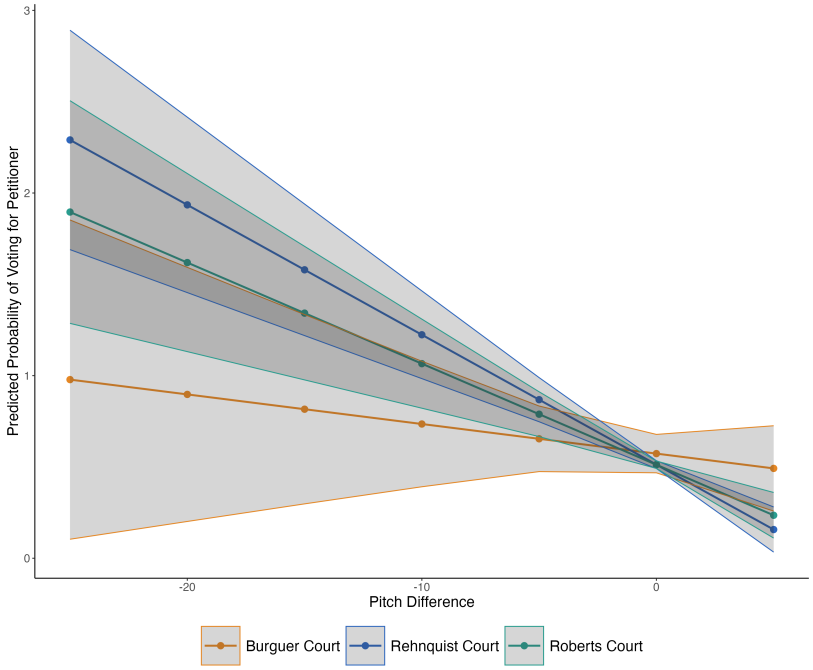
Third, amicus brief submissions supporting the petitioner are positively and significantly associated with the probability of a Justice voting in their favor, consistent with the view that external legal signals influence judicial decision-making.

Table 2: Regression Results (New controls + Interactions)

	Dependent variable:					
	Favor Petitioner Vote					
	(1)	(2)	(3)	(4)	(5)	(6)
Pitch Difference	-0.057***	-0.057***	-0.055***	-0.055***	-0.016	-0.056***
	(0.008)	(0.008)	(0.008)	(0.008)	(0.018)	(0.008)
Positive Words Difference		0.041	0.011	0.015	0.023	-0.002
		(0.160)	(0.159)	(0.159)	(0.159)	(0.160)
Amicus Brief			0.144***	0.143***	0.143***	0.143***
			(0.017)	(0.017)	(0.017)	(0.017)
Court Period: Rehnquist				-0.026	-0.060	-0.025
				(0.052)	(0.054)	(0.052)
Court Period: Roberts				-0.028	-0.061	-0.027
				(0.052)	(0.054)	(0.052)
Pitch Difference × Court Period: Rehnquist					-0.055**	
					(0.022)	
Pitch Difference × Court Period: Roberts					-0.039*	
					(0.022)	
Pitch Difference × Positive Words Difference						-0.156
						(0.124)
Constant	0.541***	0.544***	0.513***	0.540***	0.573***	0.539***
	(0.007)	(0.007)	(0.008)	(0.052)	(0.054)	(0.052)
Observations	5,209	4,977	4,977	4,977	4,977	4,977
R^2	0.011	0.010	0.024	0.024	0.026	0.025
Adjusted R^2	0.010	0.010	0.024	0.023	0.024	0.024

Note: This table shows the results from the multivariate regression estimation. The dependent variable is an indicator of whether the Supreme Court Justice voting in favor of the petitioner (1, 0 otherwise). The independent variable of interest is the Justice oral pitch difference, that is, the difference between the pitch in questions directed toward petitioners and the pitch in questions directed toward respondents. The vocal pitch was converted to standard deviations above and below the Justices' average vocal pitch. Positive Words Difference shows the difference in the proportion of positive words used when addressing the petitioner versus the respondente. Amicus Brief indicates a submission in favor of the petitioner. Court period effects included. $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$.

Figure 3: Interaction effects (Pitch Difference x Court Period)



Notes: This figure shows the interaction effects between Justices' pitch difference and court periods. Each point represents a pitch difference level and the corresponding predicted probability of voting for the petitioner. Each line represents a court period: Burguer (orange), Rehnquist (blue), and Roberts (green). N = 4,977.

4 Outlier Analysis and Threats to Validity

Q16: Based on the outlier diagnostics (Studentized residuals, leverage, Cook's distance, and DFFITS), which observations stand out as extreme? Do any specific cases appear as outliers across multiple measures?
A: There are no observations considered extreme outliers. That is, observations that are considered outliers according to all measures. There are, however, 148 cases that appear as outliers across multiple measures, but not across all measures.
Q17: How does removing outliers affect the regression results? Compare the coefficients and standard errors from the full dataset and the outlier-excluded model. Does removing outliers meaningfully change the conclusions, or are the results largely robust?
A: The removal of the outliers did not change the conclusion: higher pitcher differences are associated with a lower probability of the Justice voting in favor of the petitioner. Now, the magnitude of the estimate has changed substantially. That is, a one standard deviation increase in the Justice's pitch difference is associated with an average decrease of 8.8 percentage points in the probability of voting in favor of the petitioner, all else held constant. This is nearly a 60% change from the null dataset exercise.
Q18 (BONUS): Suppose that the measured pitch differential is related to the true difference in pitch as follows:

$$\textbf{pitch diff}_i = \delta_0 + \delta_1 \textbf{ True pitch diff}_i + u_i$$

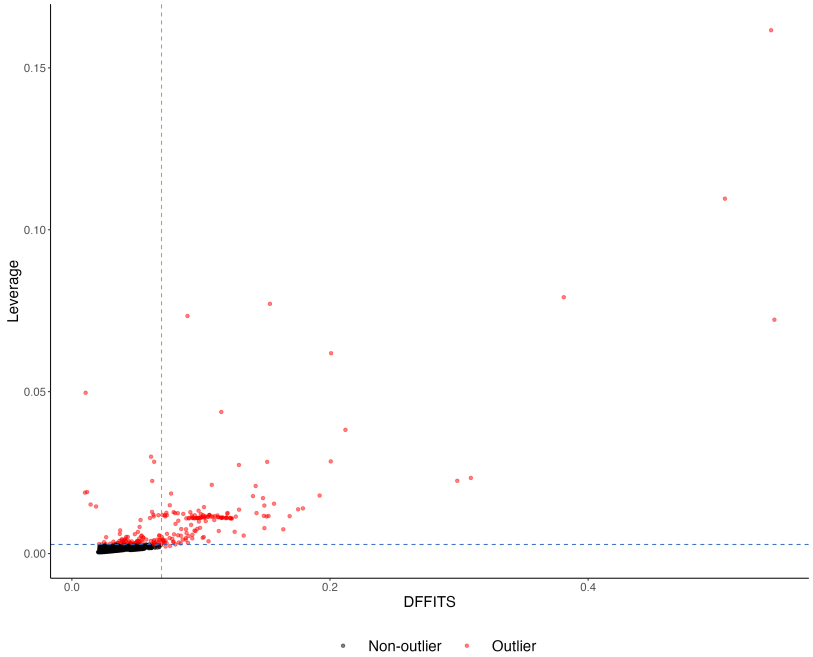
where u_i is an error term with mean zero and is independent of the true pitch differ-

Table 3: Regression Results (with and without outliers)

	Dependent variable:	
	Favor Petitioner Vote	
	(1)	(2)
Pitch Difference	-0.056***	-0.088***
	(0.008)	(0.011)
Positive Words Difference	-0.002	-0.069
	(0.160)	(0.197)
Amicus Brief	0.143***	0.144***
	(0.017)	(0.018)
Court Period: Rehnquist	-0.025	
	(0.052)	
Court Period: Roberts	-0.027	0.001
	(0.052)	(0.014)
Pitch Difference × Positive Words Difference	-0.156	-0.252
	(0.124)	(0.303)
Constant	0.539***	0.509***
	(0.052)	(0.011)
Observations	4,977	4,733
R^2	0.025	0.029
Adjusted R^2	0.024	0.028

Notes: This table shows the results from the multivariate regression estimation, with and without outliers. The dependent variable is an indicator of whether the Supreme Court Justice voting in favor of the petitioner (1, 0 otherwise). The independent variable of interest is the Justice oral pitch difference, that is, the difference between the pitch in questions directed toward petitioners and the pitch in questions directed toward respondents. The vocal pitch was converted to standard deviations above and below the Justices' average vocal pitch. Positive Words Difference shows the difference in the proportion of positive words used when addressing the petitioner versus the respondente. Amicus Brief indicates a submission in favor of the petitioner. Court period effects included only in the first model. Observations from Burguer Court were all considered outliers. *p<0.1; **p<0.05; ***p<0.01.

Figure 4: Outliers Visualization, based on DFFITS and Leverage metrics



Notes: This figure shows the potential outliers (red), based on leverage (y-axis) and DFFITS (x-axis) metrics. Orange line shows the DFFITS threshold and the blue line shows the leverage threshold. Points that are above both thresholds are unusual in X and distorts the regression line. Each point represents a Supreme Court case. N = 4,977.

ential. How does this type of measurement error affect your estimates? What would happen if u_i had a nonzero mean? What might this indicate about the accuracy of the sound-detection algorithm?

A: If u_i has mean zero and is independent of the true pitch differential, it's just white noise, and *pitch_diff*'s estimate should be centered on the true value. If the error term had a nonzero mean, this would suggest the presence of hidden factors in u_i correlated with pitch_diff_i, and the estimate would be systematically displaced from the true value. In other words, the estimate would be biased.

Q19 (BONUS): Identify two additional sources of measurement error in this study. For each one:

- **Name the type of measurement error (e.g., random error, systematic bias).**
- **Explain how it might arise in the data collection or processing.**
- **Describe its likely effect on the regression results and the interpretation of findings.**

A:

Q20 (BONUS): Beyond measurement error, what other limitations could affect the validity of this analysis? Consider potential omitted variables, issues with the study design, or alternative explanations for the observed relationships. How would you refine the research to better address the initial question of whether voice pitch predicts justices' votes?

A: I believe one fundamental problem is not accounting for the races of the Justice, petitioners and respondents. The criminal justice literature shows evidence of racial bias against minority groups. Race can affect both the probability of the Justice voting in favor of the petitioner and the pitch difference. Hence, it's a potential omitted variable bias problem.

The model could have an indicator of the petitioner's race (white x non-white) and/or an indicator of whether the Justice's race is the same as the petitioner's. There might be a problem of colinearity in this approach, so I would need to make some tests.

Overall, the above problem can be applied to other demographics, such as the petitioner's education and income. The model would need to address these confounders to have a more reliable estimate.